

Cyber Attacks

Explained

Part—9



Authentication Attacks

In the previous article, we dealt with man-in-the-middle attacks. This month, we will explore authentication attacks, which use a relatively new hacking technique capable of causing serious identity and data theft. We will also discuss how FOSS systems should be protected from such attacks.

By definition, authentication is the process of confirming identity—usually, it is a program or process that confirms the user's identity, to ensure the user really is who he or she claims to be. As we know, all computer systems that need to maintain user-specific sessions or datasets, including client-server systems and Web-enabled portals, need users to log in to some sort of authentication mechanism. Authentication is important for the overall security of software applications and systems, primarily to keep unauthorised persons from gaining access to the system and resources. Authentication isn't always about passwords; we will soon learn about other mechanisms. To understand the security challenges, we first need to understand the authentication process.

In a typical user authentication system:

1. The users try to authenticate themselves with the authentication verifier (a secure software system, which challenges the users to provide credentials such as their user ID and password).
2. Users enter the information, which is sent to an authentication module.
3. This module refers to a back-end authentication database, which contains the users' credentials; this was added when the users' profiles were created in the past. When the database entry matches the given credentials, a token is issued to the user for the active session to remove the need for authentication at each request in the session. There are also situations when a software system